

# SMART GRIDS

## From 160 to 200 GW in Two Years

Nations across the world are faced with the challenge of increasing power production while reducing the carbon footprint. They need to minimise power loss and downtime, harness alternative power sources, and so on and so forth. The numerous challenges facing them have one solution—smart grids!

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John Chambers, CEO of communications giant Cisco, has identified smart grids to be a bigger opportunity than the Internet itself! Not surprising, considering that one day the 'smart grid' will pervade the whole world. "We see a world where every significant energy-consuming device will have an IP address—be it a refrigerator, pool pump or plug-in hybrid," says Sandeep Raina, senior vice president, Cisco.

Though we are nowhere close to that vision today, it serves as a starting point for our exploration of the smart grid. A smart grid is a giant network. It is like a crochet tablecloth—complex yet simple, breathtakingly beautiful and very useful. Various points of power generation, the utility (electricity supply board) and the consumers—all connected by a network—communicate with each other, use the shared information to make intelligent decisions, and sometimes even use the network to control how each other works. Basically, if you add 'communication, intelligence and automation' to today's power distribution and transmission system,

### Intro

*A smart grid is a digital upgrade to the existing electric grid technology that has actually been quite the same for over 100 years. So what are the features of a smart grid? The USA's Department of Energy describes it quite beautifully...*

"Informed, involved, and active consumers—demand response and distributed energy resources."

"Many distributed energy resources with plug-and-play convenience; focus on renewable energy."

"Mature, well-integrated wholesale markets, growth of new electricity markets for consumers."

"Power quality is a priority with a variety of quality/price options—rapid resolution of issues."

"Greatly expanded data acquisition on grid parameters."

"Automatically detects and responds to problems—focus on prevention, minimising impact to consumer."

"Resilient to attack and natural disasters with rapid restoration capabilities."

you get a smart grid.

A Tata BP Solar spokesperson presenting an introduction to smart grids at a recent IEEE conference in Mysore rightly called it a 'digital upgrade' to current power distribution and long-distance transmission systems.

Such a system requires varied technologies and products, representing the convergence of embedded and communication technologies, electrical and electronics engineering, software and more. If orchestrated right, these technologies—collectively known as 'smart grid'—would help improve reliability of power transmission, reduce downtime, instil transparency in the system, eliminate power losses, detect and fix faults speedily, enable distributed generation of power including non-conventional power sources, and, in general, improve the experience of both the power supplier and the consumer.

There are lots of reasons to implement smart grids and lots of technologies and trends to understand before doing so. Here is a quick round-up of the same.